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Studio Session	02 (Thursday 4-6 pm)

Part A: URL of the Visualisation

Paste your visualisation URL in the empty space below.

Tip: Test your URL in incognito mode to ensure it is a working URL

<https://sgan06.github.io/Data-Visualisation-2/>

Part B: Domain Description

Describe your domain chosen for DV II

The domain chosen for Data Visualization 2 is marine biodiversity and conservation, with a specific focus on whale migration around Australia. The visualisation investigates how whale species use Australian coastal waters across space and time, with particular attention to migration timing, seasonal movement, breeding activity and long term population change. Rather than simply identifying where different whale species occur, the project explored broader behavioural and conservation questions by combining occurrence records with long term monitoring data. The main research question is: How have whale migration, breeding activity and population patterns changed around Australia?

Describe the Why and Who aspects of your DV II

The purpose of the visualisation is to help users understand how whale migration and population patterns vary across different locations, seasons and years and why long term monitoring is important for conservation. The visualisation allows users to identify spatial concentrations of whale observations, compare seasonal patterns between species, examine how observation timing changes across decades and investigate long term changes in southern right whale breeding and abundance. It is designed primarily for a general Australian audience with an interest in wildlife, marine environment and conservation, rather than for specialist marine scientists. For this reason, technical concepts are explained using accessible language, annotations and visual storytelling so

that users can understand the ecological significance of the patterns without requiring advanced statistical knowledge.

Describe the “What” aspect of your DV II

The visualisation combines data from two different real world sources. The first source is whale occurrence data from the Atlas of Living Australia, represented in the cleaned whales_for_vega.csv dataset. It contains records for humpback whales, southern right whales and blue whales, including species, observation date, year, month, day of year, season, migration phase, decade, latitude, longitude, state and observation type. These variables support the analysis of whale distribution, seasonal timing and changes in observation patterns over time. The second source is long term southern right whale monitoring data derived from the Dryad dataset associated with Australian aerial surveys from 1976 to 2024. The cleaned data includes modelled abundance for cow-calf pairs and unaccompanied whales, confidence intervals, whale abundance per 100km across four coastal survey segments and long term growth measures. Together, the two datasets provide complementary perspectives: the Atlas of Living Australia data describes where and when whales are observed, while the structured Dryad survey data provides a more appropriate basis for analysing breeding activity and long term population change.

Describe the “How” aspect of your DV II

The visualisation is implemented as a single scrolling web page using Vega-Lite, HTML and CSS, with twelve maps and charts arranged into six narrative sections. Three different map idioms are used to communicate spatial information: a point distribution map shows individual whale occurrence records, a spatially binned density map reduces overplotting and highlights areas of concentrated observations and a seasonal centroid progression map shows how the geographic centre of observations changes between migration phases without implying that individual whales were directly tracked. Temporal and behavioural patterns are then shown using a species by month heatmap, a ridgeline plot across decades and a radial seasonal chart, allowing migration timing to be examined from several complementary perspectives. Long term southern right whale monitoring is communicated using an annotated population timeline with confidence intervals, a streamgraph comparing population categories, a year by region breeding heatmap, a bump chart showing changes in the relative importance of survey segments, a connected scatterplot linking breeding and unaccompanied abundance through time and a slopegraph comparing earlier and recent breeding patterns. Interactive species selectors and informative tooltips are included where they improve understanding, while annotations and explanatory text guide the audience through the story rather than turning the page into a purely exploratory dashboard. The layout, typography, colour palette and consistent chart styling are designed to create a clear visual hierarchy and maintain a coherent marine conservation theme throughout the visualisation.

Part C: Sketch

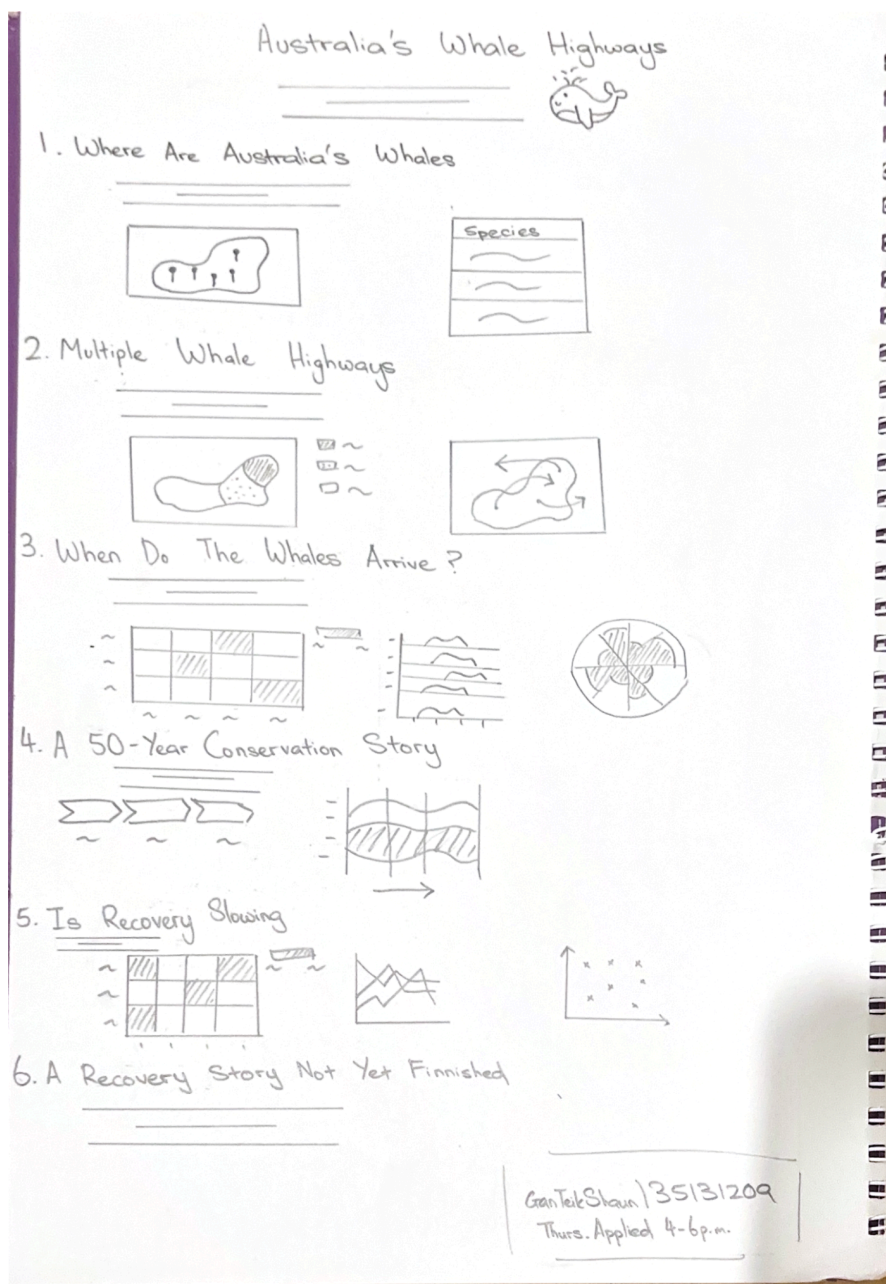
A URL to a PDF document with an image of your sketch on your GitHub account.

Tip: Test your URL in incognito mode to ensure it is a working URL

<https://github.com/SGAN06/Data-Visualisation-2/blob/8f33b7106f3f374af384a3ec66695282fb8fbfec/sketch.pdf>

Paste a screenshot of your sketch for your DV II.

Note: Use the entire page here for the screenshot and ensure that it is clearly legible and visible



Part D: Full-size screenshot of your DV II

Paste a full-size screenshot of your DV II assignment from the URL here

